

Problem

Which of the following are odd functions?

- (a) $\sin t + \cos t$
- (b) $t \sin t$
- (c) $t \ln t$
- (d) $t^3 \cos t$
- (e) $\sinh t$

Step-by-step solution

Step 1 of 5

Definition of odd function:

If a function is said to be odd only when, $f(-t) = -f(t)$.

Checking the function, $f_1(t) = \sin t + \cos t$, to be odd.

$$\begin{aligned} f_1(-t) &= \sin(-t) + \cos(-t) \\ &= -\sin t + \cos t \end{aligned}$$

Thus, the function $f_1(-t)$ is not equal to the $-f_1(t) = -(\sin t + \cos t)$.

Hence, the function $f_1(t) = \sin t + \cos t$ is not odd function.

Step 2 of 5

Checking the function, $f_2(t) = t \sin t$, to be odd.

$$\begin{aligned} f_2(-t) &= (-t) \sin(-t) \\ &= (-t)(-\sin t) \\ &= t \sin t \end{aligned}$$

Thus, the function $f_2(-t)$ is not equal to the $-f_2(t) = t \sin t$.

Hence, the function $f_2(t) = t \sin t$ is not odd function.

Step 3 of 5

Checking the function, $f_3(t) = t \ln t$, to be odd.

$$f_3(-t) = (-t) \ln(-t)$$

Thus, the function $f_3(-t)$ is not equal to the function $-f_3(t) = -(t \ln t)$.

Hence, the function $f_3(t) = t \ln t$ is not odd function.

Step 4 of 5

Checking the function, $f_4(t) = t^3 \cos t$, to be odd.

$$\begin{aligned} f_4(-t) &= (-t)^3 \cos(-t) \\ &= -t^3 \cos t \end{aligned}$$

Thus, the function $f_4(-t)$ is equal to the function $-f_4(t) = -t^3 \cos t$.

Hence, the function $f_4(t) = t^3 \cos t$ is an odd function.

Step 5 of 5

Checking the function, $f_5(t) = \sinh t$, to be odd.

$$\begin{aligned} f_5(-t) &= \sinh(-t) \\ &= -\sinh t \end{aligned}$$

Thus, the function $f_5(-t)$ is equal to the function $-f_5(t) = -\sinh t$.

Hence, the function, $f_5(t) = \sinh t$, is an odd function.

There, the d, e option functions are odd.